

JUnit

Exercise 1: Setting Up JUnit

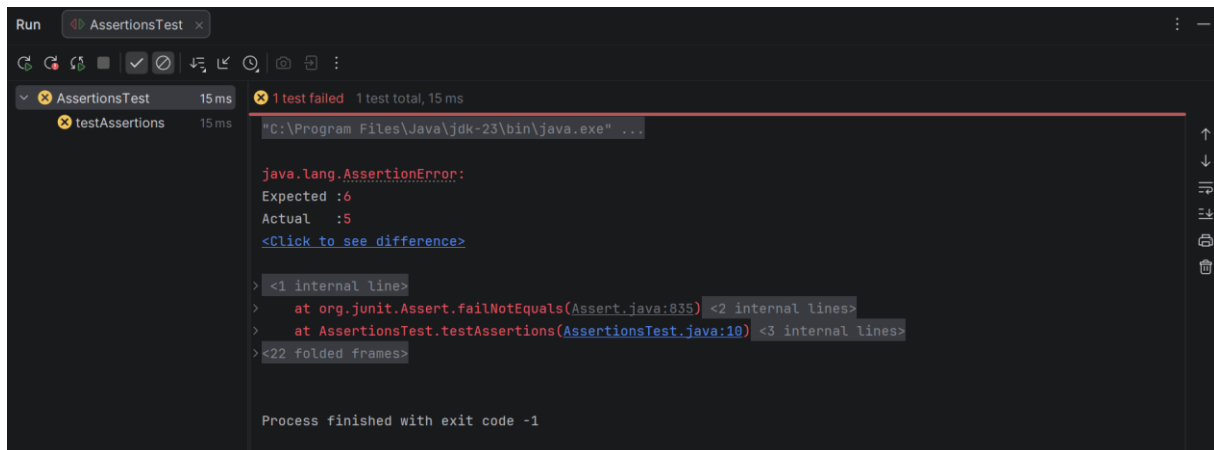


Exercise 2: Assertions in JUnit

Correct Case:



Wrong Case:



Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

Output:

===== BEFORE TEST =====

Initializing test data...

number = 5

----- ARRANGE -----

Value assigned = 5

----- ACT -----

Double of 5 = 10

----- ASSERT -----

Expected = 10

Actual = 10

Test Passed Successfully!

===== AFTER TEST =====

Cleaning up test data...

number reset to 0

=====

===== BEFORE TEST =====

Initializing test data...

number = 5

----- ARRANGE -----

Value assigned = 5

----- ACT -----

Square of 5 = 25

----- ASSERT -----

Expected = 25

Actual = 25

Test Passed Successfully!

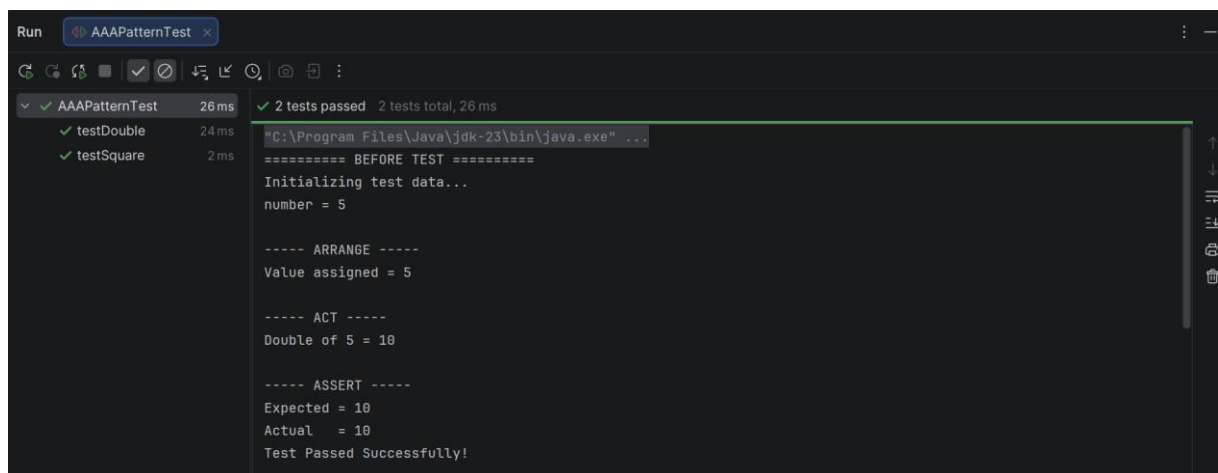
===== AFTER TEST =====

Cleaning up test data...

number reset to 0

=====

Process finished with exit code 0



The screenshot shows an IDE's Run window for a test class named AAAPatternTest. The left pane shows a list of tests: AAAPatternTest (26 ms), testDouble (24 ms), and testSquare (2 ms), all with green checkmarks indicating they passed. The right pane shows the output of the test run, which includes the following text:

```
"C:\Program Files\Java\jdk-23\bin\java.exe" ...  
===== BEFORE TEST =====  
Initializing test data...  
number = 5  
  
----- ARRANGE -----  
Value assigned = 5  
  
----- ACT -----  
Double of 5 = 10  
  
----- ASSERT -----  
Expected = 10  
Actual = 10  
Test Passed Successfully!
```